



Machine Learning-Enhanced Molecular Dynamics: Current State, Challenges and Perspectives

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Received: 22 September 2025 / Revised: 19 December 2025 / Accepted: 10 January 2026 / Published online: 24 January 2026
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Abstract

Atomic-scale simulations, such as molecular dynamics (MD), have undergone a rapid evolution during the past decades, leading to their broad adoption in various scientific and engineering fields, where they have unveiled hidden phenomena otherwise inaccessible through experiments. In recent years, the integration of machine learning (ML), either as a post-processing tool or embedded directly in a hybrid manner to the simulation workflow, has further enhanced the predictive power of MD by improving accuracy, reducing computational costs, and uncovering patterns in high-dimensional data. This review investigates the integration of MD with ML by identifying existing approaches for their combination within popular simulation platforms. It incorporates an analysis of research trends from 2020 to 2025 based on Scopus-indexed publications. By synthesizing recent developments and highlighting ongoing challenges, this review aims to guide future efforts in exploiting the full potential of MD/ML integration across scientific and engineering disciplines, such as computational chemistry and physics, biomolecular engineering, medicine, soft matter physics, materials science and nanotechnology.

1 Introduction

Molecular dynamics (MD) simulations serve as a key computational method, providing detailed understanding of how materials behave at the atomic level, especially in fields such as chemistry, materials science, and biomedical research [1, 2]. Using concepts from classical mechanics and statistical physics, MD simulations offer a time-based, detailed view of how molecules move and interact, making it possible to study reaction pathways, structural shifts, and molecular interactions that are challenging to capture in experiments [3]. Molecular dynamics have entered the literature in the late '50s [4–6], when computing power constrained applications to a few hundred atoms presented as hard spheres. It is a fact that traditional MD approaches face limitations in computational cost, timescale reach, and force field (FF) transferability [7, 8]. Simulations comprising 50000 to 100000 atoms are nowadays routine on mid-size clusters,

and trajectories close to one trillion atoms are feasible on modern supercomputers [9–13]. The leap stems from hardware acceleration (GPUs, many-core CPUs), efficient parallelization, and continued reliance on the straightforward Verlet integration scheme that lies at the heart of MD.

The core of an MD simulation is the solution of Newton's motion equations for a group of atoms, where forces are determined and atomic positions along with velocities are updated at set time intervals [14]. Managing an MD simulation also requires choosing a thermodynamic ensemble, most often the microcanonical (NVE), canonical (NVT), isothermal–isobaric (NPT), or grand canonical (μ VT) ensemble, with temperature and pressure regulated by linking the system to thermostats and barostats [15].

A popular numerical scheme widely applied in MD is the Verlet algorithm and its extensions [16]. At its core, this approach serves as an integrator for Newton's equations of motion. It has been used to study protein folding, deformation of materials, and chemical reaction processes. Verlet-based integration is the foundation of major simulation packages such as GROMACS and LAMMPS, where it supports nanosecond-scale trajectories in systems like Lennard Jones fluids. The Verlet scheme continues to be essential for both deterministic and stochastic simulations across different scales. Its time reversible, simplistic structure, further

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improved through modern variations, offers an effective balance between precision, stability, and scalability, which has kept it central to computational physics for more than fifty years [17–19].

It has been shown that MD is capable of providing precise estimates for important transport properties such as the diffusion coefficient, shear viscosity, and thermal conductivity across a range of materials and fluids, including water and electrolyte solutions. [20, 21] Selecting between Equilibrium Molecular Dynamics (EMD) and Non-Equilibrium Molecular Dynamics (NEMD), along with ensuring the precision of the interatomic potentials or force fields (FFs), plays a major role in how reliable the predictions are [22–24]. EMD and NEMD reflect two fundamentally different ways of determining transport coefficients. In EMD, the focus is on studying the spontaneous variations of a system in thermodynamic equilibrium, whereas NEMD applies an outside disturbance to create a steady state away from equilibrium, enabling direct measurement of the transport property. The decision between the two approaches is guided by the property being examined, the characteristics of the system, and the preferred trade-off between computational expense and precision [25].

In addition to the basic EMD and NEMD methods, there are advanced techniques used to better understand transport processes. One such method is *ab-initio* MD (AIMD), which combines quantum mechanical calculations with classical molecular dynamics to follow how atomic systems change over time [26, 27]. Unlike conventional molecular dynamics that use predefined force fields, AIMD determines atomic forces in real time from electronic structure calculations, most often through Density Functional Theory (DFT) [28]. This method enables AIMD to reliably represent complex processes such as the breaking and forming of chemical bonds, movement of charges, and electronic excitations, without needing any predefined interatomic potentials [29]. By removing the need for pre-set FFs, this method can simulate chemical reactions, but its very high computational demands limit it to relatively small systems and brief simulation periods [30].

To address sampling constraints and broaden the exploration of complex energy landscapes, Monte Carlo (MC) approaches are often combined with MD. Monte Carlo methods are particularly effective at drawing representative configurations from a statistical ensemble, which makes them valuable for examining equilibrium behavior. Within MD, certain MC techniques are employed to improve sampling and system control, helping simulations navigate configurational space more efficiently, cross energy barriers, and achieve better performance [31–33].

A key issue in establishing an MD simulation framework is the incorporation of interatomic potentials and FFs,

which define how atoms and molecules interact, guiding their motion and overall behavior. How well MD performs depends strongly on the quality of the interatomic potential, such as the Lennard-Jones (LJ) model, or established FFs like AMBER, CHARMM, OPLS, and GROMOS, which calculate forces between atoms and shape the potential energy surface of the system [34]. Research on simple fluids usually applies simplified models such as the square-well or triangular-well potentials. Since these have abrupt changes that standard MD integrators cannot handle directly, they are typically converted into smooth mathematical forms, such as the continuous square well potential or the constant force method [35].

Of importance are also the software platforms that implement numerical calculations. It has been observed that open-source packages such as LAMMPS and GROMACS dominate academic use due to their modular architectures and compatibility with external machine learning (ML) frameworks, while commercial platforms like Materials Studio maintain strong presence in industrial design pipelines [36]. Quantum and electronic structure codes (Gaussian, ORCA) continue to underpin high-accuracy reference datasets, and Quantum ESPRESSO remains essential for large-scale DFT and ML-accelerated materials screening. A broader convergence across disciplines is also evident. Subject area distributions indicate growing overlap between chemical engineering, condensed matter physics, and computational materials science, while new techniques that combine MD with machine learning (ML) or deep learning (DL) (or other artificial intelligence (AI) methods) dive into domains previously dominated by empirical models [37–43].

More specifically, over the past two decades, MD simulations have evolved from rigid, force-field-driven calculations to highly flexible, data-enhanced frameworks integrating ML. ML offers powerful solutions to these bottlenecks, enabling accelerated simulation, improved predictive power, and deeper physical insights. Moreover, challenges associated with the interconnection of these methods are discussed. Data transferability, active learning, and uncertainty quantification remain open problems, especially for fluids under extreme conditions where experimental benchmarks are sparse. Recent work on scalable equivariant potentials and active learning frameworks aims to close this gap [44], and the challenge is to connect it with physics-aware potentials of atomistic accuracy [45]. Advances in high-performance computing, including multilevel parallelism [46], are equally important, enabling hybrid simulations that merge quantum accuracy with macroscopic time and length scales.

This review investigates the intersection of ML and MD, providing a comprehensive survey of approaches having

been developed during the past years [47–51]. In the Sections that follow, we try to synthesize these developments, mapping the intersection of MD and ML under the prism of chemistry, physics, and materials science, mainly. By reviewing various software platforms, methodological innovations, and publication trends, we aim to establish the current state of the field and identify opportunities for future research in data-driven molecular simulation.

2 ML Concepts Combined with MD

2.1 Machine-Learning Interatomic Potentials

The integration of ML into MD and quantum chemistry frameworks represents a pivotal shift in atomistic simulations, as the computational cost of ab-initio methods like DFT poses significant barriers. Although DFT provides accurate potential energy surfaces (PES), which are crucial for the reliability of any MD simulation, this cost restricts the system under simulation in short timescales and modest system sizes. This form makes them incapable of modelling complex quantum phenomena and chemically diverse environments. This practical limitation has given rise to the use of empirical potentials, which provide fast access to energies and forces at the expense of ab-initio accuracy. In such a way, after the dominance of AI/ML techniques in most scientific principles, the main idea for the representation of PES has come from fitting large data sets from electronic structure calculations into an ML algorithm.

Machine-Learning Interatomic Potentials (MLIPs) are constructed as surrogate models and connect the precision of ab-initio methods with the speed of classical empirical FFs, marking a major step forward in computational materials science and chemistry. They achieve near-DFT accuracy at a computational cost much closer to that of classical FFs. Their main idea is to represent the forces and energies in a system using ML approaches [52, 53]. An MLIP operates on the same input descriptors as DFT, i.e., the spatial coordinates and atomic numbers of the constituent atoms, supplemented by spin or charge information where relevant. The model's primary output is the total potential.

All these characteristics highlight the advantages of MLIPs over classical FFs. While classical FFs operate on a fixed topology, they cannot model the breaking or forming of chemical bonds. Moreover, rare events at long time scales cannot be revealed with direct MD [54]. Reactive force fields (e.g., ReaxFF) exist but are complex to parameterize and lack the general accuracy of MLIPs. An MLIP learns the entire PES from DFT and is able to handle bond dissociation and formation with quantum-level accuracy. In terms of transferability, an MLIP trained on a diverse dataset, can

accurately model a material across a wide range of phases and environments. It can successfully handle crystalline defects, grain boundaries, amorphous structures, and surfaces [55]. In such a way, when an MD simulation employs an accurate MLIP, the derived properties are reliable [56].

Among various ML methods, artificial neural networks (ANNs) and, especially, deep NNs (DNNs) are the most widely used [57]. In general they function well, acting as universal approximators, learning the complex and nonlinear links between an atom's surroundings and its contribution to the total energy [58]. Nonetheless, their computational cost typically remains one to two orders of magnitude higher than that of the simplest empirical FFs, but their parallelization is possible in current HPC and GPU architectures, to allow them exhibit near-linear scaling with system size [59]. These models reproduce energies with high accuracy, but they do not employ an explicit physical functional form; instead, they learn the shape of the potential energy surface directly from the training data.

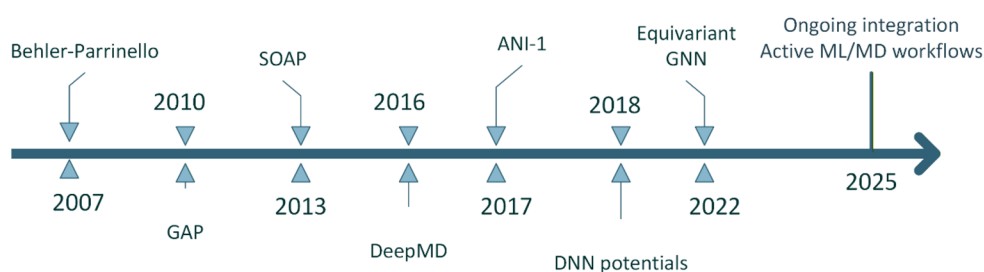
The integration of ML and MD has progressed through several significant milestones, as seen in Fig. 1. In 2007, NN potentials were first introduced by Behler and Parrinello [60], representing high-dimensional potential energy surfaces. An automated framework for generating finite-range interatomic potentials directly from quantum-mechanical forces and energies without explicit electrons has been presented in Bartók et al [61]. Test applications to bulk semiconductors, iron, and graphene [62], among others, have shown that the proposed Gaussian Approximation Potential (GAP) reproduces the ab initio PES with remarkable accuracy, at a fraction of the computational cost. Gaussian Approximation Potentials use Gaussian process regression (GPR), a Bayesian nonparametric method. This framework provides predictions along with uncertainty estimates and often achieves high accuracy. The Spectral Neighbor Analysis Potential (SNAP) is based on GAP, but the main difference lies in the use of a linear dependence of the energy on the bispectrum elements [63].

Another aspect closely connected to the integration of ML and atomistic simulations is how to map atomic information into a numerical array that can be fed into a ML algorithm. A distance metric with superior properties for constructing PES has been introduced, the Smooth Overlap of Atomic Positions (SOAP) [64], while some extensions have been also developed [65]. Schütt et al (2017) [66] introduced deep tensor NNs (DTNNs), which can accurately represent chemical environments from pairwise distances.

The years that follow, the Accurate Neural network engine for Molecular Energies (ANAKIN-ME or ANI-1), a deep NN potential designed to achieve high accuracy and transferability for organic molecules by learning from DFT calculations, was proposed [67]. ANI-1 core innovation is a

Fig. 1 Milestones of molecular dynamics and Machine learning interconnection, from the introduction of Behler-parinello NN potentials to the current trend of active, integrated ML/MD methods

Year	Milestone
2006	Behler–Parrinello Neural Network Potentials
2010	Gaussian Approximation Potentials (GAP)
2013	SOAP descriptors
2016	Deep Potential/DeepMD framework
2017	ANI-1 Neural Network Potentials
2018	Advanced DNN potentials (e.g., SchNet, HIP-NN)
2022	Equivariant GNN potentials (e.g., NequIP, MACE, PaiNN)
2025	Ongoing integration / Active ML–MD workflows



novel molecular representation built from a highly modified version of Behler-Parrinello symmetry functions, which construct single-atom atomic environment vectors (AEVs). This AEV framework allows training of the NN across combined configurational and conformational space, providing a scale of transferability.

After that, SchNet, a variant of DTNNs that employs a deep convolutional neural network (CNN), has driven the attention from molecular descriptors to the network architecture, learning molecular representations directly from first-principles data [68]. It is able to capture complex PESs while remaining computationally fast. The resulting model is interpretable, physics-aware, and recovers fundamental chemical knowledge from its learned atomic embeddings. Other schemes based on deep networks are the Hierarchically Interacting Particle Neural Network (HIP-NN) [69], PhysNet [70], and the recent Message Passing Network with Iterative Charge Equilibration (MPNICE) [71].

GNNs have also shown superior performance in learning interatomic potentials capturing local atomic environments and their connectivity [72, 73]. Owing to their fast inference, robust generalization, and scalability, GNNs have managed to advance various chemical applications, from ab-initio simulations and reaction kinetics to synthesis planning and drug discovery [74]. For instance, the Neural Equivariant Interatomic Potential (NequIP) is a DL approach utilizing the power of GNNs, exhibiting exceptional data efficiency

even from sparse data [75]. MACE is an advanced message passing GNN that employs higher body order messages and achieves increased computational efficiency [76]. The UMA architecture is also an implementation based on equivariant GNNs which has introduced critical modifications that enable it to efficiently scale and to accept additional descriptors, including total charge, spin state, and the target DFT functional settings, thereby expanding its capacity as a surrogate model [77].

Moment Tensor Potentials (MTPs) form another family of MLIPs that relies on systematic polynomial expansion. Studies report that MTPs balance accuracy and computational cost effectively. They are often used together with active learning. In active learning, the potential is updated during the simulation by finding atomic configurations where the model is uncertain and then adding new quantum calculations for those cases. This concentrates computation on the most informative configurations and makes training more efficient [78]. An exemplary case is provided by Zeni et al(2021), who developed ML-derived force-fields for Au nanoparticles and applied unsupervised characterization of atomic environments to accurately simulate melting, showing that the transition initiates at the nanoparticle surface [79].

Nowadays, from 2020 onward, the adoption of hybrid MD-ML frameworks has become a reality [53] and this is expected to be the new standard [80]. This historical path

highlights the transition from isolated algorithmic improvements to full-scale integration of ML models into active MD workflows.

2.2 Coarse-Grained MD with ML Integration

Several ML techniques are currently shaping the next generation of MD simulations, posing as alternatives to traditional empirical strategies and computationally demanding simulations [81–83]. Coarse graining (CG) is a technique often used to atomistic modelling of complex molecules, such as polymers and biomolecules, which group single atoms to nodes with the same physical behavior, significantly reducing computational costs. In the review of Ye et al [84], the involvement of ML in CG methods is summarized to four critical areas: regulating the CG process, constructing adequate descriptors, generating representative training data, and optimizing the loss function.

Machine learning CG potentials have been widely incorporated in protein dynamics investigation, where a general understanding of system dynamics is ambiguous [85]. A physics-based CG model is presented in [86], which is able to simulate various protein systems and can be extrapolated for similar systems. Various ML methods have been used in many ways to identify and parameterize interactions or mapping functions between the detailed atom structure and the simplified nodes representations. Supervised learning models like DNN's or unsupervised learning models like Principal Component Analysis (PCA) are employed for the construction of various geometries. Hybrid CG/ML frameworks were proposed in evolving traditional methods, from shape-based CG (SBCG) to shape-based CG 2 (SBCG2) [87, 88]. Another example includes CGnets, i.e., a NN class that respects physical invariances (e.g., rotation, translation) by learning a free energy (potential of mean force) and deriving forces from its gradient [89]. Three different models for water, i.e., ML-BOP, ML-BOP_{dih}, and ML_{mW} have been applied to investigate the mesoscale mechanism of ice grain formation and growth from supercooled water and are suggested for studying complex problems like phase transitions, nucleation, and interfacial properties. These models will enable simulations at system sizes and timescales that are currently inaccessible to popular potentials like the TIP4P and TIP5P models [90].

2.3 ML as Post-Processing and Analysis Tool

Having generated extensive MD trajectories, the subsequent step involves mining this data for significant structural and dynamical insights. To this direction, ML methods have been widely applied to the prediction of properties in fields such as materials science, chemistry, physics, and engineering, by

automatically identifying complex patterns and correlations that cannot be reached through conventional data analysis. This kind of surrogate modeling enables rapid prediction of key system properties, bypassing the need for full atomistic simulations. These innovations collectively reduce computational cost while improving generalizability [47, 91].

For instance, by relying on data-driven models, ML has been employed to study catalytic processes, where it estimates energetics, activity, and selectivity in reactions such as oxidative coupling of methane and asymmetric catalysis, as well as in experimental chemistry, where it helps predict solubility and optical rotation, reducing the need for extensive laboratory testing [92]. In crystalline and solid-state systems, ML models such as GNNs capture structural relationships to predict formation energies, band gaps, and phonon density of states, offering guidance in the search for materials with favorable thermal or superconducting behavior [93, 94]. In the energy sector, ML supports the development of metal-ion batteries by aiding electrode selection, electrolyte formulation, and monitoring of operating conditions, with successful applications in refining electrode compositions using X-ray diffraction data [95, 96]. In construction materials and composites, predictive models estimate thermal, mechanical, and optical properties of systems such as cement-based materials, aerogels, and phase-change compounds [97]. In metallurgy, ML is used to evaluate strength, ductility, and microstructure-sensitive properties of alloys, complementing or reducing experimental work [98, 99].

Beyond specific applications, ML contributes to the exploration of a wide range of materials, including perovskites, high-entropy alloys, superconductors, and catalysts. It has been adopted for phase diagram construction, prediction of service behavior such as fatigue and corrosion, and estimation of microscopic quantities including lattice constants and structural energies [100–103]. Several recent reviews deal with materials property prediction, such as magnetic materials [104], graphene [105], alloy materials [106], Metal-Organic-Frameworks (MOFs) [107], and solid-state electrolytes [108], among others. Overall, ML is becoming an essential part of materials research, providing tools that integrate with experiments and simulations to improve understanding and guide the design of advanced materials. Table 1 presents a selection of representative ML methods and their applications within the MD/ML framework.

Simple methods like Linear Regression (LR) have their role on MD/ML synergy and nanoscale property prediction, especially when paired with advanced structural descriptors, offering a computationally cheaper alternative to complex NNs [72, 109]. Decision trees (DCs) and their more accurate ensemble forms have an important but different role in the ML workflow for materials science. Rather than

Table 1 A list of commonly employed ML algorithms for MD speed-up and property prediction (illustrative, not exhaustive)

ML Category	ML Technique	Primary Role	Use Cases	Refs.
Regression & Statistical Models	Linear Regression	Computing forces & energies, property prediction.	Computationally cheap, NN alternative	[72, 109]
	Sparse/Symbolic Regression	Discovering governing physical equations from data	Identify the most relevant features for parsimonious models, Calculate macroscale properties from nanoscale data	[73, 110–112]
Ensemble & Tree-Based Methods	Decision Trees, Random Forest, Gradient Boosting	Classification & Regression for material properties	Classify crystal structures (e.g., perovskites), determining influential chemical descriptors	[113–116],
Neural Networks (NNs)	DNNs, CNNs, RNNs	Developing ML-based FFs, predicting quantum-mechanical energies	Core architectures for learning complex PES from quantum mechanics data	[53, 117]
	GNNs	Developing ML-based FFs, representing molecular systems	Naturally models atomic connectivity. Effective for long-range interactions, adaptable to diverse chemistries.	[118–120],
Reinforcement & Adaptive Learning	RL	Adaptive sampling, exploring rare events & conformational transitions	Agents trained to explore rare events (e.g., protein folding pathways)	[47, 121]
	Self-Supervised & Transfer Learning	Training models with limited labeled data	Reduces dependency on large quantum-mechanical datasets	[47, 121]
Dimensionality Reduction	Principal Component Analysis	Simplifying & analyzing high-dimensional simulation data	Extracts essential features from complex trajectories for analysis and visualization	[73, 110].

computing forces and energies for MD, these models are mainly used for classification and regression to estimate material properties or pinpoint key factors in synthesis. For instance, approaches such as Random Forest, Gradient Boosting and its advanced form (XGBoost) have been employed to calculate the strain-induced crystallization (SIC) of natural rubber [114], separate perovskite from non-perovskite structures, determine the main chemical descriptors influencing the synthesis of materials like zeolite [113], and calculate tensile properties (e.g., elongation) for Ti (titanium) and high entropy quaternary alloys (HEA), composed of Fe (Iron), Ni (Nickel), Cr (Chromium), Co (Cobalt), and Cu (Copper) [115, 116].

Various types of NNs such as Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs), which are instrumental in developing ML-based FFs, have been proven capable of predicting quantum-mechanical energies, and analyzing simulation trajectories [53]. Graph neural networks (GNNs), which naturally depict molecules and materials as graphs, enabling them to learn from the built-in connectivity and symmetry of atomic arrangements, are gaining popularity. They are particularly effective for representing long range interactions and adapting to a wide variety of chemical settings [118].

Other notable methods include Reinforcement Learning (RL) for adaptive sampling and for exploring rare conformational transitions [47, 121], various dimensionality reduction techniques like PCA for simplifying high-dimensional data, and sparse regression methods like LASSO and symbolic regression for discovering governing physical equations from data [73, 110–112]. RL agents have been trained to explore rare events, such as ligand unbinding or protein folding [121]. Self-Supervised and transfer learning have also emerged. These approaches reduce dependency on large labeled datasets, making ML accessible for limited-data MD studies [47, 121]. Collectively, these methods extend the power of MD/ML to a broader range of scientific problems and materials systems.

The t-distributed Stochastic Neighbor Embedding (t-SNE) [122] is a ML technique that maps a large group of high-dimensional objects, to reduce its dimensions to a smaller two-dimensional group, in such a way that doesn't affect structural neighboring relationships. During an MD simulation, where data exists in a massive configurational space (defined by all the atom positions, bonds, angles, etc.), t-SNE can be employed to map this space through dimensionality reduction. The method presents some limitations, such as in single cell RNA-seq data, where it doesn't preserve the global geometry of the objects, making it highly sensitive to random initialization, which doesn't

come with agreement with biologically important hierarchical structures. To equalize this drawback, PCA initialization techniques coupled with fast interpolation-based t-SNE (Flt-SNE) [123] are applied [124].

Furthermore, DL methods like autoencoders (AE's) have been proven effective at discovering the underlying collective variables of a system by learning a compressed representation (latent space) of the atomic configurations [125]. For instance, by exploiting AEs denoising capabilities, an AE network has been trained to derive the radial distribution function (RDF) from a single snapshot of an LJ liquid to calculate the average RDF [126]. Recently, an ML framework (IceCoder) that combines a variational AE (VAE) with the SOAP descriptor has been presented [127]. It compresses high-dimensional SOAP vectors into a 2-D latent space, allowing for clear visualization and distinction between different phases.

3 A Survey of Popular Atomistic Simulation Platforms

This section provides a survey of computational tools that form the backbone of modern computational science and engineering. These platforms can be broadly categorized into three groups: Classical MD engines, quantum chemistry and electronic structure tools, and visualization and analysis software, which are critical for interpreting the complex, high-dimensional data produced by these simulations. Figure 2 presents a cloud-form list. In what follows, classical MD engines will be discussed, outlining their primary capabilities and typical use cases to guide researchers in selecting the appropriate tool for relevant scientific and engineering test cases. Emphasis is given on the tools' ability to integrate with ML. Table 2 lists some of the most

popular MD engines along with the type of ML integration, applications which employ them and possible limitations. More details will be given on the dedicated Sections for each tool that follow.

3.1 LAMMPS

Large-scale Atomic/Molecular Massively Parallel Simulator was first introduced in 1995 and has since become a cornerstone of modern MD, renowned for its versatility, efficiency, and extensive FF library [177]. A recent flagship methodology paper by Thompson et al [128] provides the definitive overview of LAMMPS capabilities, parallel scalability, and support for ML potentials. Specialized workflow tools including OpenMSCG for bottom-up coarse-grained model generation [178] and Twister for constructing twisted 2D heterostructures [179] further demonstrate how LAMMPS is actively evolving to serve data-driven pipelines. A concrete demonstration of the hybrid paradigm is the work of Salarnia et al (2023) [180], who combined LAMMPS nanoparticle simulations with ML regression to predict effective thermal conductivity showcasing how physics-based data and ML surrogates can accelerate property estimation.

Building on this foundation, LAMMPS has become a central platform for integrating MLIPs with MD. Table 3 provides an overview of ML integration techniques with LAMMPS. Important milestones include the introduction of physics-informed neural network (PINN) models [181], rapid artificial neural network (RANN) potentials [131], SNAP [132], and DeePMD [129]. Collectively, these developments have allowed LAMMPS to host a growing ecosystem of ML potentials that can be tailored to different classes of materials and simulation conditions.

Fig. 2 Popular platforms and software tools used for MD simulations, quantum chemistry and DFT simulations, and analysis/visualization

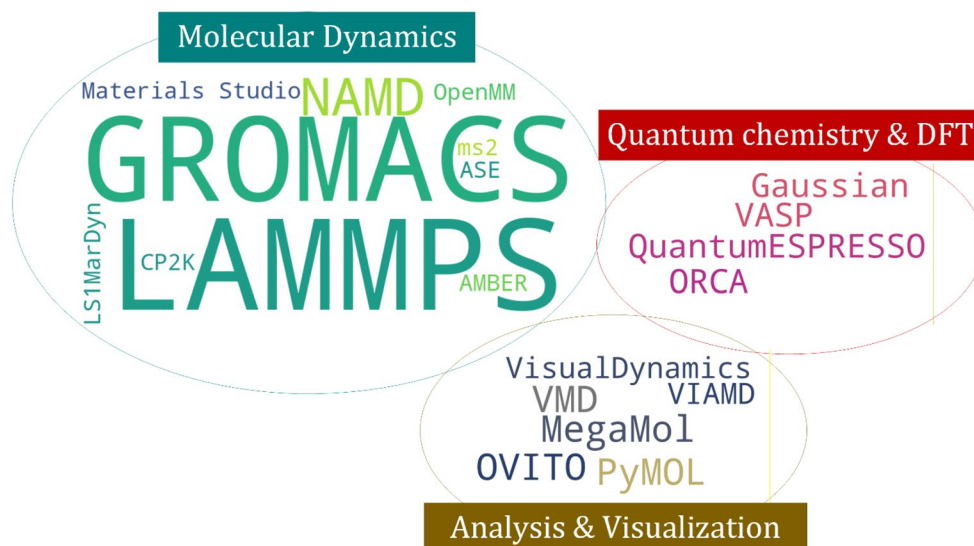


Table 2 MD softwares and how they embed ML into their framework, with application examples and current limitations

Software	ML Integration	Applications	Limitations	Ref.
LAMMPS	DeePMD, FitSNAP, PINN, SNAP	Thermal conductivity, diffusion, defect dynamics, ionic liquids	Few direct transport validations. Coupling with visualization still growing	[128–141]
GROMACS	TorchMD, TorchMD-Net, GNNs	Thermal transport, ionic/proton diffusion, biomolecular transport	Limited benchmarking vs. experiments. ML transport studies sparse	[142–146]
NAMD	Colvars (RL), DeepDriveMD (DNNs)	Free-energy landscapes, adaptive sampling, ensemble simulations	ML features still emerging, primarily biomolecular focus	[147–151]
Materials Studio	COMPASS II,III	Polymers, composites, adsorption, corrosion studies	Proprietary, limited direct ML coupling (beyond COMPASS)	[152–156]
OpenMM	PyTorch plugins, Deep Potential, MACE	Water/ionic liquids, polymers, hybrid QM/MM	Smaller user base in materials vs. biomolecules	[157–164]
CP2K	DeePMD, DMFF, GNN potentials	Liquids, ionic liquids, catalytic surfaces	Performance depends on backend. ML interfaces still maturing	[91, 165, 166]
AMBER	Deep potential MD	Ionic liquids, polymer–solvent systems, transport properties	Historically biomolecular, limited ML adoption	[167–170]
ASE	NequIP, Dscribe descriptors	Ionic diffusion, conductivity	Depends on external DFT backends (VASP, GPAW, QE)	[171–173]
LS1MarDyn	Hypothetical ML integration proposed	Fluids, mixtures, large-scale HPC studies	Limited ML implementations to date	[174–176]

Table 3 LAMMPS milestones and applications

Category	Milestone/Application	Significance	Refs.
Early MLIP Integration	PINN	An early framework that merged physical constraints with deep learning to accurately predict forces, particularly for metallic systems.	[130, 181]
Major MLIP Frameworks	DeePMD, RANN, SNAP	Widely adopted MLIPs that interface with LAMMPS. They are trained on quantum-level data to run massive simulations (millions of atoms) with high accuracy.	[129, 131, 132]
Thermal Transport	MLIPs for Thermal Conductivity	Using MLIPs (like DeePMD or ReaxFF-nn) with Green-Kubo formalism to accurately compute thermal properties in complex materials (e.g., MOFs, molten salts, graphene).	[129, 133–135]
Defect & Mechanics	PINN & SNAP for Alloys/Ceramics	Predicting elastic properties, surface energies, and defect dynamics. This is crucial for understanding transport in materials like high-entropy alloys.	[182–184]
Hybrid Strategies	Surrogate Modeling & Active Learning	Using ML to predict properties from simulation. Also, using active learning to refine MLIPs ‘on-the-fly’ during a simulation.	[180]
Integration Tools	MLMOD & FitSNAP	Essential ‘bridge’ software. MLMOD connects LAMMPS to external ML libraries. FitSNAP streamlines the entire process of creating and validating SNAP potentials.	[137, 138]
Data-Driven Workflows	OpenMSCG & Twister	Specialized tools that show LAMMPS evolving into a data-driven platform, used for generating models (coarse-graining) or building complex structures.	[178, 179]

Applications of NN network potentials have been employed to compute the thermal conductivity of metal-organic frameworks such as CALF20, combining Green-Kubo formalisms with ML force evaluations [133], ML-enhanced reactive FFs have been used to capture thermal transport in graphene and carbon nanotubes while

maintaining reactive chemistry fidelity [134]. Large-scale benchmarks using DeePMD within LAMMPS have extended these capabilities to noble metal alloys, enabling direct simulation of phonon transport in systems exceeding millions of atoms [135], along with simulations of water and ionic liquids with high accuracy, providing thermal

conductivity predictions for molten salts and electrolytes [129]. Furthermore, PINN and SNAP frameworks have been adapted for high-entropy alloys and multicomponent ceramics, where accurate defect energetics are essential for modeling radiation-induced transport. Although such works do not always present transport coefficients explicitly, the resulting datasets often feed into multiscale models that bridge atomic diffusion with continuum-scale conductivity [182–184].

Parallel to MLIP development, workflow tools and modular packages have expanded LAMMPSs accessibility for machine-learning studies. The MLMOD interface, for instance, provides a general bridge between LAMMPS and various ML libraries, allowing researchers to couple external neural network models without extensive code modification [137]. FitSNAP offers a streamlined pipeline for generating and validating SNAP-based ML potentials, supporting descriptor calculation, training, and integration with LAMMPS runs [138]. These frameworks have been instrumental in promoting reproducibility and standardization of ML workflows, particularly when scaling up to high-performance computing environments.

3.2 GROMACS

GROMACS remains at the forefront of high-performance computing through GPU/CPU co-processing strategies [185] and the multilevel parallelism architecture [46]. Table 4 provides an overview of recent popular ML integration

techniques with GROMACS. Deep Learning potentials can now be interfaced via TorchMD [186], offering interfaces for equivariant NN potentials that respect rotational symmetries and are well-suited for complex molecular systems. Workflow automation is supported by CHAPERONg [3], while the recent development of gmxapi, a Python native interface, has opened opportunities for seamless integration of ML models and custom plugins directly into GROMACS workflows [142].

GAP-based studies were performed for the modeling of anisotropic thermal conductivity in $\beta\text{Ga}_2\text{O}_3$, an oxide semiconductor relevant for high-power electronics [193]. Similar ML-based approaches have been applied to crystalline and amorphous silicon, where DL potentials capture phonon transport behavior more accurately than empirical models [194]. These efforts highlight how MLIPs can improve predictions of heat transport by incorporating anharmonic interactions and complex defect dynamics that classical potentials may miss. More recent work employing GNNs, such as MACE and Allegro models, has further expanded the scope of ML-enabled GROMACS simulations to systems where directionality and local environment variations strongly influence transport [186].

Another promising trend is the development of interpretable and computationally efficient ML potentials, which aim to bridge the gap between accuracy and cost. Recent spline-based models have achieved near state-of-the-art performance while maintaining high throughput in MD engines [196]. Their compatibility with GROMACS offers a path

Table 4 GROMACS milestones and applications

Category	Milestone/Application	Significance	Refs.
ML Integration Frameworks	TorchMD/TorchMD-Net	An interface that connects GROMACS to PyTorch, enabling the use of modern Neural Network (NN) potentials, including “equivariant” NNs that respect molecular symmetries	[186]
Python API	gmxapi	Python interface designed to make GROMACS easier to integrate custom ML models, plugins, and complex workflows	[142]
Biomolecular	SARS-CoV-2 Dynamics, Proton Hopping, Ligand Diffusion	Demonstrated the use of ML-guided ensemble simulations at an exascale level to study virus spike dynamics. Study quantum-level transport processes (like proton movement) in biological systems, such as enzyme active sites	[187–192]
Thermal Transport	GAP & DL Potentials	Using MLIPs like Gaussian Approximation Potentials (GAP) to model complex heat transport (e.g., in $\beta\text{-Ga}_2\text{O}_3$ or silicon) more accurately than classical models	[193, 194]
Ionic Transport	Neural IL FF	A specific ML potential designed to predict diffusion and conductivity in ionic liquids and proton transfer systems, where classical FFs often fail	[195]
Emerging MLIPs	GNNs (MACE, Allegro)	Integrating advanced GNNs to handle systems where local atomic environments and directionality are critical	[186]
Workflow & Efficiency	CHAPERONg, Spline-Models	Tools to automate simulation workflows [3]. New, efficient “spline-based” models are being developed for high throughput without sacrificing accuracy [131]	[3, 196]

toward practical ML-enhanced simulations of transport properties in large, realistic systems such as nanofluids or biomembranes. As these models mature, they are expected to reduce the barrier to adopting ML-driven FFs in routine simulations, especially for research groups with limited access to specialized hardware.

3.3 OpenMM

OpenMM is a widely adopted open-source toolkit designed for high-performance MD simulations, known for its GPU-optimized architecture and flexibility in force-field implementation. The recent version of OpenMM 8 introduced direct compatibility with ML potentials, enabling seamless integration with frameworks such as PyTorch [158]. This upscale has broadened OpenMM's applicability beyond biomolecular systems, suggesting it as a robust tool for materials and fluids research. Milestones and applications are summarized in Table 5.

Ding and Huang developed an OpenMM plugin for the Deep Potential energy representation, validating it through simulations of bulk water and demonstrating improvements in accuracy for structural and thermodynamic properties [159]. Complementary work by Spahr et al employed Deep Potential Molecular Dynamics (DPMD) to evaluate viscosity and self-diffusion of water, showcasing strong agreement with experimental data [160]. Jung and Yethiraj combined OpenMM-based MD with DNNs to analyze poly(ethylene oxide) in room-temperature ionic liquids, providing insight into miscibility and structural transitions [162]. More recently, Zhao et al introduced MACE-MP-0, a high-accuracy molecular potential for materials modeling, and demonstrated its integration into OpenMM workflows for efficient simulations of diverse condensed-phase

systems [163]. These advances extend the platform's reach to polymeric and hybrid materials, where both accuracy and computational efficiency are essential.

Efforts to merge OpenMM with broader ML workflows are currently evolving. The OpenMM-Python-Force extension [164] allows user-defined ML potentials to operate natively within MD simulations, while OpenMM-MiMiC provides a framework for hybrid QM/MM simulations relevant to nanofluidic systems [197]. Together, these developments indicate a growing synergy between data-driven force-fields and classical MD engines.

3.4 CP2K

CP2K is an open-source program for atomistic simulations that supports both classical and quantum MD, making it widely used for materials and fluids research. Its modular design and the Quickstep algorithm provide efficient DFT and hybrid functional calculations [91]. Performance benchmarks have highlighted CP2K's scalability on modern architectures, including GPUs, enabling large-scale ab-initio MD studies [165]. This dual capability, with handling quantum-level accuracy and classical speed, makes CP2K a versatile platform for investigating condensed-phase phenomena across polymers, ionic liquids, and catalytic materials (Table 6).

Mo et al presented an ML-accelerated MD approach integrated with CP2K, achieving accurate thermodynamic predictions for condensed-phase systems [166]. Sivakumar et al extended these capabilities to catalytic interfaces, using active learning and CP2K to describe adsorbate dynamics on metal surfaces [139]. Polymer-ionic liquid interactions have also been explored, such as combined CP2K-based simulations with DNN analysis to map phase behavior in

Table 5 OpenMM milestones and applications

Category	Milestone/Application	Significance	Refs
Key ML Milestone	OpenMM 8	A major release that introduced direct compatibility with ML potentials, enabling seamless integration with frameworks like PyTorch	[158]
MLIP Integration	DeePMD Plugin	An extension to run Deep Potential Molecular Dynamics (DPMD) within OpenMM, leveraging its GPU performance for ML-driven simulations.	[159]
Fluids	DPMD for Water Properties	Using the DeePMD plugin to accurately simulate bulk water, validating its structural properties and predicting transport coefficients (viscosity, self-diffusion).	[160]
Polymers	Polymers & Ionic Liquids	Combining OpenMM with DNNs to analyze complex systems (e.g., PEO in ionic liquids)	[162]
Materials	MACE-MP-0 Potential	Integrating a high-accuracy, next-generation ML potential (MACE) for efficient and accurate simulation of diverse condensed-phase materials	[163]
Customization Tools	OpenMM-Python-Force	An extension that allows users to implement their own custom-defined ML potentials directly in Python, offering maximum flexibility	[164]
Hybrid Methods	OpenMM-MiMiC	A framework for integrating hybrid QM/MM simulations, relevant for nanofluidic systems	[197]

Table 6 CP2K milestones and applications

Category	Milestone/Application	Significance	Refs.
Hybrid Quantum/ML	ML-Accelerated MD/Active Learning	CP2K AIMD engine to generate data on-the-fly to train ML models.	[139, 166].
MLIP Integration	DeePMD plugin, MACE-MP-0, Equivariant GNNs	Direct integration of modern, high-accuracy MLIPs, allowing them to run within the CP2K framework	[75, 129, 163].
Advanced ML Framework	DMFF	Differentiable FFs to blend physics with ML	[198]
Complex Fluids	Polymer & Ionic Liquid Analysis	Simulate complex multi-component systems where quantum-level polarization effects are critical	[162]

poly(ethylene oxide)/ionic liquid mixtures, highlighting the platform's flexibility in complex multi-component systems [162].

Integration of explicit ML frameworks has advanced CP2K's functionality beyond traditional *ab-initio* MD. MLIPs such as MACE-MP0 have been interfaced with CP2K to improve accuracy and efficiency in modeling catalytic and energy materials [163]. The DeePMD plugin [129] allows direct use of DNN potentials trained on high-level quantum data, enabling accurate simulations of liquids and surfaces at reduced computational cost. The DMFF framework further enhances this ecosystem by introducing differentiable FFs compatible with CP2K workflows [198]. Additionally, equivariant GNN potentials have been demonstrated within CP2K-based studies, achieving near *ab-initio* accuracy for large-scale condensed-phase systems [75]. These developments indicate a clear trajectory toward hybrid quantum ML methods that combine the advantage of first-principles modeling with the scalability of data-driven approaches.

3.5 AMBER

AMBER is a long-standing MD package widely known for its biomolecular applications but increasingly employed in studies of fluids, ionic liquids, and polymer-solvent systems. Its development of the general AMBER FF (GAFF) has established a path for accurate predictions of thermodynamic and transport properties in ionic liquids [168], while later efforts focused on performance optimization, including GPU-accelerated simulations using particle mesh Ewald electrostatics, enabling microsecond-scale trajectories [167]. These foundational capabilities underpin recent applications of AMBER in condensed-phase materials research.

Force-fields for bio-polymers in ionic liquids have been developed, studying cellulose behavior in EMImOAc/water environments to elucidate structural and solvation effects [199]. Extensions to charged materials systems were also demonstrated [200], presenting predictive modeling of ionic conductivity and dielectric properties, while hydrogen bond

dynamics in binary ionic liquid mixtures have been investigated, providing insight into ion-pair mobility and interactions [201]. Recent efforts are now integrating ML with AMBER workflows to improve accuracy and efficiency in condensed-phase simulations. DPMD and MACE GNN potentials are employed to model ionic liquids, comparing their predictions against AMBER-based baselines [202]. The transferability of ML-derived interatomic potentials for ionic liquid simulations within AMBER-compatible frameworks is assessed, highlighting improved fidelity in transport properties [141].

All these studies indicate a shift in AMBER's role from a traditional biomolecular MD tool toward broader materials and fluid applications, with ML potentials such as DPMD and MACE driving this transition. The combination of established force-fields like GAFF with emerging ML-based frameworks allow AMBER to pose as a flexible tool for probing transport properties and structural behavior in ionic liquids and polymer-solvent systems. A summary of all the above is given in Table 7. In the future, it is expected that continued integration of ML potentials and GPU-accelerated algorithms is likely to enhance AMBER's utility in high-throughput property prediction and hybrid reactive–nonreactive simulations for materials science.

3.6 NAMD

NAMD has recently entered the ML era with the release of NAMD 3, which delivers GPU kernels and task-based parallelism for petascale performance [148]. Reinforcement-learning protocols implemented through the Colvars module enable adaptive exploration of free-energy landscapes, while the DeepDriveMD workflow links DNNs to NAMD for real-time steering on supercomputers [203].

The program's emphasis on adaptability has also encouraged the development of various sampling strategies that precede its current ML-related features. Methods such as accelerated molecular dynamics (aMD) [149] and generalized multi-copy algorithms [150] demonstrate how NAMD has long supported large-scale ensembles and flexible simulation workflows. Taken together, these capabilities show

Table 7 AMBER milestones and applications

Category	Milestone/Application	Significance	Refs
Foundational FF	GAFF (General AMBER FF)	A well-established classical force field enabling accurate study of non-bio systems (e.g., ionic liquids). It now serves as a crucial baseline for ML potential validation.	[168]
Classical Applications	Ionic Liquids & Polymers	Used as a reliable tool for studying complex fluids, such as cellulose in ionic liquids, ionic conductivity, and H-bond dynamics	[199–201]
ML Integration	DPMD & MACE Potentials	Adopting modern MLIPs (DeePMD, GNNs) to model complex systems (like ionic liquids) where classical FFs may be limited.	[202]
ML Application	Benchmarking & Transport Properties	Assessing the transferability and accuracy of MLIPs (like DPMD, MACE) against AMBER's classical models to validate their improved fidelity for transport properties.	[141, 202]

Table 8 NAMD milestones and applications

Category	Milestone/Application	Significance	Refs.
Core Engine	NAMD 3	A major release delivering petascale performance on new architectures through GPU kernels and task-based parallelism.	[148]
ML-Driven Sampling	Reinforcement Learning (RL) via Colvars	Implements RL protocols directly within the simulation, enabling adaptive exploration of free-energy landscapes.	[203]
ML-Workflow Integration	DeepDriveMD	A workflow that links Deep Neural Networks (DNNs) to NAMD, enabling real-time steering and control of simulations running on supercomputers.	[203]
Foundational Strength	Advanced Sampling (aMD, etc.)	NAMDs long-standing support for flexible, large-scale sampling methods (like accelerated MD) provides the foundation for its new ML-driven adaptive workflows.	[149, 150]

the evolution of NAMD from its original parallel implementation [147, 151] into a present-day, ML-assisted platform. Ongoing work positions NAMD not only as a powerful MD package but also as a versatile framework for merging physics-based models with data-driven tools, uniting advanced sampling, reinforcement learning, and interactive simulation control. A summary is given in Table 8.

3.7 Other Tools and Wrappers

Materials Studio (MS) is a widely adopted commercial platform for atomistic modeling, particularly for simulations of polymers, composites, and porous materials. Recent research illustrates its versatility in modeling diverse systems and incorporating advanced FFs such as COMPASS II and the more recent COMPASS III, which is optimized through ML fitting to quantum mechanical datasets. Bio-composite systems have been approached using COMPASS II FFs within MS to assess interfacial adhesion and elastic behavior, notably in polypropylene–cellulose composites [155].

The Atomic Simulation Environment (ASE) is a Python framework that facilitates atomistic simulations by providing a unified interface to multiple electronic structure and classical simulation codes. It is fundamentally different from the other packages, since it is not a simulation engine

itself but rather a powerful Python-based wrapper [172]. Its primary function is to provide a single, unified interface that allows a user to script and control many different simulation codes.

IS1MarDyn is a highly scalable MD simulation code optimized for massively parallel execution on supercomputing architectures. IS1MarDyn incorporates an efficient dynamic load balancing scheme to ensure high scalability across many processors. Designed to simulate large systems containing over four trillion particles, it enables the application of pair potentials to study systems at previously inaccessible length and time scales, making it suitable for applications that were previously computationally prohibitive [174].

ms2 is a molecular simulation tool developed to calculate thermodynamic properties, following the computational trends, utilizing HPC techniques like Hybrid MPI and OpenMP [204]. At its latest version release, the Mie and the Tang-Toennies potential have been added, further increasing the tools capability for the extraction of thermodynamic data. The Kirkwood-Buff integrals, based on radial distribution functions, were introduced, while an alternative to the Green-Kubo formalism was presented using Einstein relations at sampling transport properties, making the tool suitable for Nucleation and Vapor mixed simulations. [205]

3.8 Post-Processing and Visualization Tools

Effective post-processing analysis and visualization are critical components in MD/ML workflows. Visualization tools allow us to analyze trajectories, monitor structural changes, extract features, and create meaningful representations. Several specialized software packages have emerged, offering capabilities from 3D playback to trajectory clustering and automated feature extraction.

PyMOL is a widely adopted molecular visualization platform originally developed by DeLano for 3D structural rendering and analysis of chemical and biological systems [206]. Recent developments illustrate PyMOL's evolving role in ML-assisted pipelines. For example, the PyVibMS plugin enables visualization of vibrational modes derived from quantum or ML-generated FFs, facilitating interpretation of transport-relevant vibrational spectra in molecular and solid systems [207]. MegaMol is an open-source visualization framework developed to support the post-processing of complex simulations and experiments. In high-performance computing contexts, the framework supports detailed simulations of biological systems, including biophysics-driven models such as those representing skeletal muscle structure and function [208]. OVITO is a widely used software for visualization and post-processing of atomistic simulation data, with embedded computational options, such as Wigner-Seitz defect analysis, grain boundary identification, clustering of atomic environments, and calculation of quantities such as radial distribution functions and mean squared displacements.

Visual Dynamics is a browser-based application designed to simplify setup, execution, and visualization of molecular dynamics simulations using GROMACS [209]. Visual Molecular Dynamics (VMD) is extensively used for visualizing and analyzing molecular dynamics trajectories. It is a standard tool for inspecting diffusion pathways, validating ionic mobility results, and preparing trajectory data for further analysis. Its broad adoption and plugin ecosystem make VMD valuable in both classical and ML-assisted simulations involving transport phenomena. Visual Interactive Analysis of Molecular Dynamics (VIAMD) is an interactive visualization framework designed for detailed inspection of MD trajectories [210].

4 Publication Trends in MD-ML Software (2020–2025)

To assess the recent evolution of ML-enhanced classical MD tools, we have conducted a bibliometric analysis using publication data indexed in the Scopus database. This methodology provided an objective, quantitative foundation for

identifying trends in software adoption, scientific output, and interdisciplinary relevance between 2020 (January) and 2025 (November).

The search (Date: November 2025) was focused on articles published between 2020 and 2025. The search query was designed to retrieve all publications that contained three essential concepts: (1) the term “*Molecular Dynamics*”, (2) the tool term (e.g. “*LAMMPS*”, “*GROMACS*” etc), and (3) at least one term from a comprehensive list of machine learning identifiers such as “*Machine Learning*”, “*regression*”, “*Deep Learning*”, “*Artificial Intelligence*”, “*Neural Networks*”, “*Data-Driven*”, “*GNN*”, or “*Graph Neural Network*”. This strategy ensured the retrieved articles were precisely focused on the integration of ML techniques with a specific MD engine. Publications were also filtered by type to retain only peer-reviewed journal articles, conference papers, and scientific reviews. We also note that if a related paper does not explicitly refer to the simulation tool used, then it cannot be counted by our Scopus search.

The publication landscape for MD software tools integrated with ML shows marked growth and diversification over the 2020–2025 period. Figures 3 and 4 summarize cumulative publication counts per software package, annual publication rate and key dimensions of subject area distribution trend. Collectively, these data provide insight into the adoption dynamics of both commercial and open-source platforms, the disciplinary breadth of MD/ML research, and the factors influencing software selection for specific application domains.

The bar chart in Fig. 3 accounts the annual publication output for each MD simulation package. A general observation is the exponential growth of the field, with the total number of publications increasing dramatically each year. The most significant trend is the rise of LAMMPS (light green) and GROMACS (dark green), with LAMMPS having the sharper growth, accelerating from a strong position in 2020 to become the leader in annual publications by 2025. GROMACS follows a similar growth, establishing itself as the clear second-most-published platform currently. AMBER (pink) remains a major contributor, accounting for a large volume of publications, but appears to stabilize or slightly decline in 2025.

In a parallel growth, OpenMM (magenta) shows the steepest relative expansion, growing from a very small base in 2020 to a significant player in 2025, while ASE (brown/ochre) shows similarly impressive, steady growth, solidifying its niche and growing from ~30 to ~140 publications. NAMD (light blue) shows a small, linear increase in publications. Its growth is not keeping pace with the rest of the field, suggesting its market share is shrinking relative to the rapidly expanding platforms.

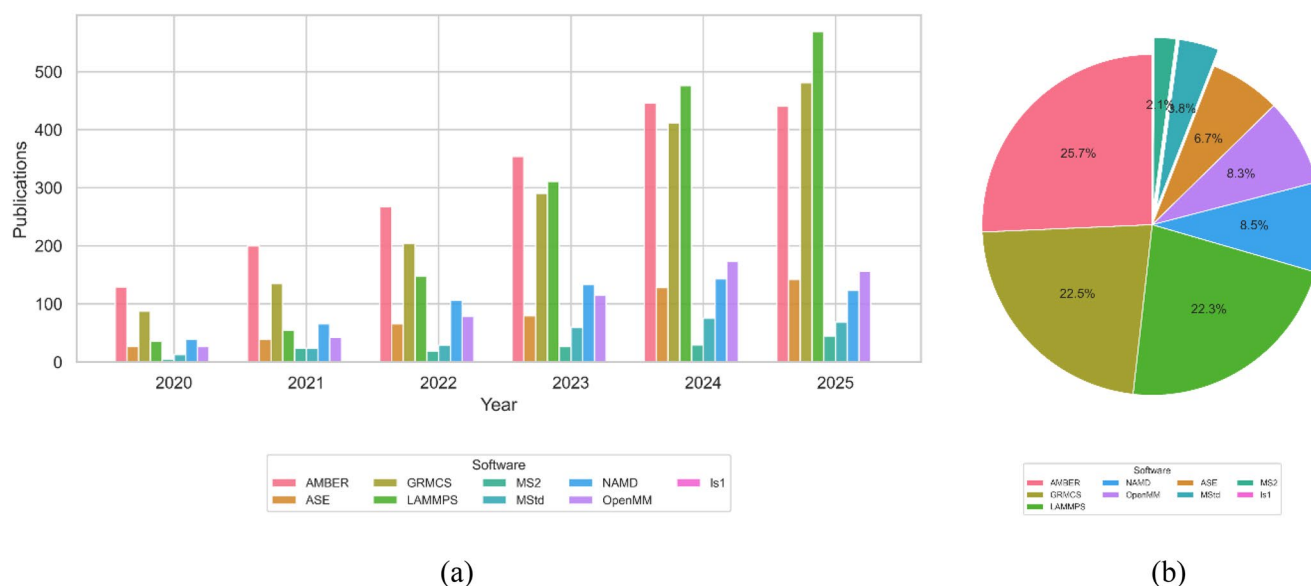


Fig. 3 Popular MD platforms from 2020 to 2025. a) annual publication counts per year, and b) cumulative publication counts per platform

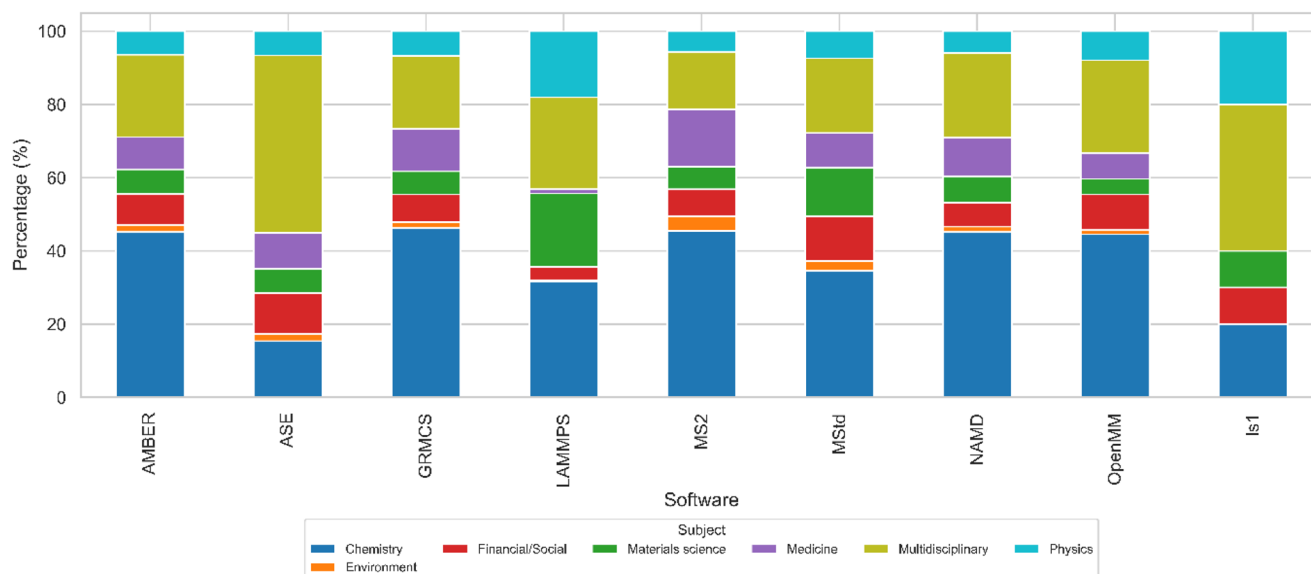


Fig. 4 Subject area publication counts per MD software platform

On the other hand, it is important to place the low publication counts for platforms like Materials Studio, ms2, and ls1MarDyn into the proper context. These numbers do not necessarily indicate a lack of utility but rather reflect different use cases and reporting conventions. For instance, ms2 and ls1MarDyn are high-performance, specialized codes, developed to solve specific, complex problems (e.g., large-scale hydrodynamics, complex fluid mixtures) and are not intended as general-purpose, all-in-one platforms like LAMMPS or GROMACS. Their lower citation counts are somehow expected. Moreover, Materials Studio is a commercial, GUI-driven environment, used as a front-end to

setup, run, and analyse simulations, and researchers may omit to mention its use.

The pie chart in Fig. 3 illustrates the total cumulative publication share for each software package over 2020–2025. The field is clearly dominated by three main packages that account for over 70% of all publications, AMBER (25.7%), GROMACS (22.3%), and LAMMPS (22.5%). The second group accounts for much of the remaining share, all holding similar cumulative totals, i.e., OpenMM (8.7%), ASE (8.5%), and NAMD (8.3%). The remaining packages account for the final ~4% of the literature.

Figure 4 shows the subject of publications for each software package, normalized to percentage share. It is of interest to note that Multidisciplinary (including engineering, mathematics, and computer engineering) and Chemistry-related research show high shares across most platforms, reflecting the integration of ML-driven techniques that bridge materials, chemistry, and data science. We note here that the Financial/Social category refers to papers that, along with the atomistic simulation, they also present a financial analysis for the industrial sector, pointing out how these methods might have an impact on economy.

The first cluster, including AMBER, GROMACS, OpenMM, and NAMD, is dominated by Chemistry publications, signifying their primary use in biochemistry and biomolecular simulation. These tools also contribute to Multidisciplinary and Medicine research. On the other hand, LAMMPS stands out for Chemistry, Materials and Physics, with a large share in Multidisciplinary research. The ASE package displays a unique profile, being overwhelmingly dominated by Multidisciplinary publications, which reflects its function as a Python-based wrapper that integrates various other codes. Materials studio exhibit a mixed profile similar to LAMMPS, balancing Chemistry, Materials Science, and Multidisciplinary work, ms2 is closer to GROMACS style, and ls1MarDyn shows the strongest specialization in Physics, underscoring its focus on statistical mechanics and fluid dynamics.

5 MD/ML Challenges

Despite its widespread success in studying atomic-scale phenomena, traditional MD is constrained by high computational demands, especially for long timescale simulations. Force-field accuracy is another limitation; while empirical potentials are computationally efficient, they may lack generalizability across chemical environments. These limitations have given rise to research efforts to enhance MD with aid of ML.

It is a fact that MD simulations face built-in challenges related to hardware, software, and methodological aspects. These constraints influence the precision, scalability, and usefulness of MD in fields such as biomedicine, materials science, and quantum chemistry. A structured overview from recent studies highlights several key issues. Time scale barriers exist, and classical MD has difficulty in capturing rare events like protein folding or chemical reactions that unfold on micro- to milli-second timescales. Hardware acceleration can lengthen accessible trajectories, but progress is still limited by energy demands and inefficiencies in parallel execution [2, 211, 212].

To anticipate hardware constraints, CPU and GPU clusters are widely employed. Nonetheless, they often experience reduced efficiency in strong scaling because of communication delays and memory bottlenecks. For instance, CHARMM simulations typically achieve less than 80% efficiency beyond 64 nodes [213, 214]. Specialized accelerators, provide high computational throughput but require specific design and generally lack versatility across diverse FFs [215]. Software and algorithmic hurdles are also present, albeit ML is employed. Potentials developed with ML extend coverage but demand extensive training sets and may produce errors when applied outside known configurations.

On the other hand, classical MD tends to under sample rare states, while biasing methods like meta dynamics may distort energy surfaces. Incorporating uncertainty-aware ML helps address this but adds extra computational cost [216]. Finally, the multiscale modeling option, such as linking MD with continuum approaches (i.e., such as computational fluid dynamics), still remains unsystematic and can introduce artifacts at the boundaries between scales [217].

Machine learning itself has also many challenges to overcome, usually related to data preparation, training, and interpretation. Reliable training data typically require costly quantum-mechanical calculations, creating a bottleneck for large-scale studies [218]. Datasets are often imbalanced, under-representing rare events such as high-energy conformations, which leads to biased predictions [219]. Also, the high dimensionality of molecular data further demands careful feature selection to avoid overfitting, and models must respect physical invariances to ensure physically consistent outcomes [53, 72, 219].

Furthermore, training of ML models adds another layer of complexity. Deep neural networks and GNNs demand significant computational resources and specialized hardware, with training often hindered by non-convex loss landscapes that trap algorithms in local minima [72, 220]. Hyperparameter optimization is time-consuming, empirical, and sensitive to initialization, making models prone to overfitting and limited generalization [72, 218]. Moreover, while models may perform well for a specific molecular system, achieving transferability across different chemistries or thermodynamic conditions remains a key challenge [53]. Training data must be also produced for surfaces, not only for bulk systems [221].

Interpretability is a further limitation. Complex ML models may provide accurate predictions while obscuring their reasoning, a “black box” issue that hinders the extraction of mechanistic insights and reduces confidence in results, particularly in applications such as drug development and materials design [219, 220, 222]. Approaches to improve

interpretability exist but often add computational cost and require specialized expertise.

6 Perspectives

By analyzing the data extracted from our literature search during the past years, it is revealed that ML integration into MD platforms is transitioning from rare to mainstream across multiple scientific regions. While commercial tools are dominant in volume, open-source packages are driving methodological innovation, cross-disciplinary adoption and are central to pioneering ML frameworks. Anticipated developments such as scalable equivariant NNs [186] and interpretable spline-based potentials [196] are likely to further increase publication output on these fields.

Efforts to integrate ML into MD are increasingly focused on overcoming barriers related to data, efficiency, and interpretability. Active learning protocols will have a main role in reducing data-generation costs by selectively identifying the most informative training samples, while transfer learning and multifidelity data fusion will combine high-accuracy datasets with larger, low-cost sources to improve model robustness [218, 223]. Physics-informed models, which embed physical constraints into ML architectures, will further improve generalization and reduce data demands [224]. A recent example employs atomistic detail to upscale coarse-grained simulations with physics-aware methods [225]. In hardware terms, parallelization and optimized GPU-based implementations will also aid toward scale expansion of ML-driven MD, making simulations of systems exceeding 100 million atoms with near ab-initio accuracy a standard process.

Future developments are also expected to emphasize interpretability, with new algorithms and visualization tools designed to clarify ML predictions. Symbolic regression approaches, which provide an explainable solution over usual “black-box” models may also find its role here [226]. Strengthening collaboration between computational scientists (from materials, physics, biology, and chemistry, mainly), computer scientists and engineers will also be crucial for tailoring ML approaches to MD. Together, these advances will establish ML methods to become more reliable and versatile tools for future molecular simulation designs [227, 228].

7 Conclusion

The integration of machine learning with molecular dynamics has become a reality during the past years. This review has been focused on understanding current state and

limitations. Through an extended literature review from Scopus sources, a foundational framework to guide future research in the field has been provided.

It has been argued that classical software platforms have been upscaled in this direction and high-performance computing has made it possible. The landscape is characterized by the dominance of feature-rich, established platforms like Gaussian for generating high-fidelity training data with ab-initio accuracy, and open-source codes like GROMACS and LAMMPS, which are central to pioneering next-generation neural network potentials and active learning frameworks. In such a way, powerful hardware and flexible software are working synergistically. This allows us to run more detailed and larger molecular simulations than ever before. This shift has made data-driven, predictive computing the new standard in science, in fields from biomolecular interactions and solid-state properties to emerging areas like ionic liquids.

We have also discussed key methodological challenges that continue to hinder broader application of MD/ML frameworks, especially for predicting properties under extreme states. The analysis confirms that the path forward is contingent upon concurrent progress in two areas: (1) the development of more transferable and uncertainty-aware ML models, and (2) their tight integration with increasingly powerful computing architectures.

In summary, the integration of machine learning into molecular dynamics will, eventually, mature into a mainstream application and transform computational science, where possible. Scalability, efficiency, and interpretability are the key aspects of this process. The next frontier lies in moving beyond “black-box” models through advances in physics-informed architectures and explainable approaches, which will provide critical insights and robustness in ML-driven predictions.

Author Contributions CS: Data curation, Formal analysis, Writing – original draft, Visualization. FS: Conceptualization, Writing – review & editing, Visualization. TEK: Writing – review & editing.

Funding Open access funding provided by HEAL-Link Greece.

Data Availability The data used for this review came from Scopus and is included within the article.

Conflict of Interest The authors have no conflicts to disclose.

Competing Interests The authors declare no competing interests.

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